Arnau Muñoz Barrera

Computer Engineer | Data Science & AI

Barcelona, Spain

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*Computer Engineer and Master's in Data Science student with hands-on experience transforming complex, high-frequency data into actionable insights. Experienced in building **data pipelines**, scalable backend services, and **interactive dashboards** to improve system visibility and support operational decisions. Interested in applying analytical problem-solving and technical expertise to **business intelligence**, reporting, and automation in an international environment.*

EXPERIENCE

Energiot Devices S.L. **Barcelona, Spain**
Software Engineer *May 2026 – Present*

- Design, develop, and maintain **APIs and backend data services** for IoT infrastructure, supporting secure ingestion, synchronization, validation, and processing of **high-frequency telemetry data** from connected devices.
- Build analytical **dashboards, monitoring platforms, and geospatial visualizations** that transform operational data into actionable insights for infrastructure supervision and **data-driven decision-making**.
- Optimize cloud-based **data pipelines, database persistence, and processing workflows** to support growing data volumes, improving scalability, performance, consistency, and fault tolerance.
- Communicate directly with **customers and stakeholders** to understand technical requirements, clarify business needs, discuss solutions, and provide updates on implementation progress and issues.
- Coordinate technical projects and feature delivery by planning tasks, prioritizing requirements, aligning implementation with stakeholder expectations, and collaborating with the team to meet deadlines and project objectives.
- Take ownership of architecture, integrations, and backend improvements while strengthening **production monitoring, debugging, and deployment workflows** to improve platform reliability and maintainability.

Energiot Devices S.L. **Barcelona, Spain**
Full Stack Engineer Intern *February 2025 – May 2026*

- Implemented and maintained **REST APIs, backend services, and frontend features** for IoT applications, connecting device data collection and processing with dashboards and interactive maps.
- Assisted with the development, testing, and debugging of full-stack functionality, investigating issues across application layers and helping deliver reliable features in a production environment.
- Worked with databases and data-processing workflows to improve data access, persistence, and application performance, contributing to more reliable and scalable services.
- Collaborated with senior engineers on feature implementation, code reviews, bug fixes, and technical improvements, gaining experience with version control, maintainable code, and the software development lifecycle.

TECHNICAL SKILLS

- **Data & Languages:** Python, SQL, Pandas, ETL Pipelines
- **Cloud & Backend:** AWS, Docker, REST APIs, Git

RECENT PROJECTS

- **Hackathon Winner – UABTheHack 2025** Repository
- **Hackathon UPC 2026** Repository
- **ML Model – Car Price Prediction** Repository

ACADEMIC FORMATION

UNIVERSITAT POLITÈCNICA DE CATALUNYA (UPC) **Barcelona, Spain**
Master's in Data Science *2026 – 2028*

UNIVERSITAT AUTÒNOMA DE BARCELONA (UAB) **Barcelona, Spain**
Bachelor's in Computer Engineering *2022 – 2026*

LANGUAGES

- English: **C1 Certified**
- Native: **Spanish and Catalan**
- Beginner: German